

MDL → Answer → Key

$$H(A) = \sum -p_i \log_2(p_i) = -\left(\frac{6}{11} \log_2 \frac{6}{11} + \frac{3}{11} \log_2 \frac{3}{11} + \frac{1}{11} \log_2 \frac{1}{11} + \frac{1}{11} \log_2 \frac{1}{11}\right) = 1.61$$

weather!

Cinema, Tennis, Tennis, Tennis

$$H(S_{\text{sunny}}) = -\left(\frac{1}{4} \log_2 \frac{1}{4} + \frac{3}{4} \log_2 \frac{3}{4} + 0 + 0\right) = 0.811$$

Cinema Cinema
Cinema
shopping

$$H(S_{\text{windy}}) = -\left(\frac{1}{4} \log_2 \frac{1}{4} + \frac{3}{4} \log_2 \frac{3}{4} + 0 + 0\right) = 0.811$$

Cinema Cinema
stay in

$$H(S_{\text{rainy}}) = -\left(\frac{2}{3} \log_2 \frac{2}{3} + \frac{1}{3} \log_2 \frac{1}{3} + 0 + 0\right) = 0.918$$

$$H(S|A = \text{weather}) = \frac{4}{11} \cdot 0.811 + \frac{4}{11} \cdot 0.811 + \frac{3}{11} \cdot 0.918 = 0.84$$

$$I_4 = 0.776$$

A = Parents

Cinema, Cinema, Cinema, Cinema, Cinema

visit = 5

$$H(S_{\text{visit}}) = -1 \log_2 1 + 0 \log_2 0 = 0$$

No visit = 6

$$H(S_{\text{no visit}}) = -\left(\frac{3}{6} \log_2 \frac{3}{6} + \frac{3}{6} \times \frac{1}{6} \log_2 \frac{1}{6}\right) = 1.792$$

Shopping, Tennis, Tennis, Shopping, Cinema

$$I_4 = 1.6 - \frac{6}{11} \times 1.792 = 0.633$$

A = Cash

Cinema → 3

Tennis → 2

stay → 1

shopping → 1

Rich = 7

Poor = 4

$$H(S_{\text{rich}}) = -\left(\frac{3}{7} \log_2 \frac{3}{7} + \frac{2}{7} \log_2 \frac{2}{7} + 2 \times \frac{1}{7} \log_2 \frac{1}{7}\right) = 1.842$$

$$H(S_{\text{poor}}) = -\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4}$$

Cinema → 3

Tennis → 1

$$= 0.811$$

$$I_4 = 1.6 - \frac{7}{11} \times 1.842 - \frac{4}{11} \times 0.811 = 0.149$$

A = cinema

yes = 5

no = 6

Cinema → 3

shopping → 1

Tennis → 1

$$H(S_{\text{yes}}) = -\left(\frac{3}{5} \log_2 \frac{3}{5} + \frac{1}{5} \log_2 \frac{1}{5} + \frac{1}{5} \log_2 \frac{1}{5}\right) = 1.391$$

Cinema = 3

stay = 1

Tennis = 2

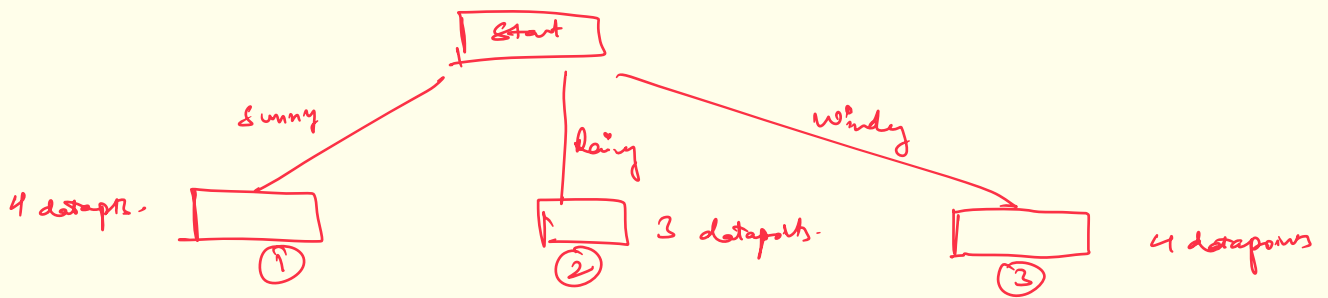
$$H(S_{\text{no}}) = -\left(\frac{2}{6} \log_2 \frac{2}{6} + \frac{1}{6} \log_2 \frac{1}{6} + \frac{3}{6} \log_2 \frac{1}{6}\right)$$

$$= 1.459$$

$$I_4 = 1.616 - \sum_{i=1}^3 \frac{1}{11} \times 1.331 - \frac{6}{11} \times 1.459$$

$$= 0.197$$

max information gain is through decision = weather.
1st decision is weather.



now apply algorithm for node (1)

$$H(S) = \text{Cinema} \rightarrow 1 \quad \text{Tennis} \rightarrow 3$$

$$+ \left(\frac{1}{4} \log\left(\frac{1}{4}\right) + \frac{3}{4} \log\left(\frac{3}{4}\right) \right)$$

now for attribute parent:

$$\text{Visit: } H(S_{\text{visit}}) = \text{Cinema} \rightarrow 1$$

$$= \frac{1}{4} \log 1 + 0 \log 0 \dots$$

$$= 0$$

$$H(S_{\text{not-visit}}) = \text{Tennis} \rightarrow 3$$

$$= \frac{3}{3} \log 1 + 0 \log 0 \dots$$

$$= 0$$

I_4 is max for parent.

similarity for (2) attribute is parent.

For Node ②

$$H(S = \text{Parent}) = \frac{2}{4} \times 0 + \frac{1}{2} \times 1 = \frac{1}{2}$$

$$H(S = \text{Cash}) = \frac{3}{4} \times 0.92 = 0.69$$

$$H(S = \text{Sunny}) = \frac{2}{4} \times 0 + \frac{1}{2} \times 1 = \frac{1}{2}$$

2nd decision for Node ② can be cash or parent both.

Similarly 3rd decision for node ③ can be calculated until we get pure Nodes.

Visit $\rightarrow 2$

Not visit $\rightarrow 2$

Rich $\rightarrow 3$

Poor $\rightarrow 1$

$$H(S_{\text{visit}}) = -\frac{2}{2} \log \frac{2}{2} + 0 + 0 = 0$$

$$H(S_{\text{not-visit}}) = -\frac{1}{2} \log \frac{1}{2} - \frac{1}{2} \log \frac{1}{2} = 1$$

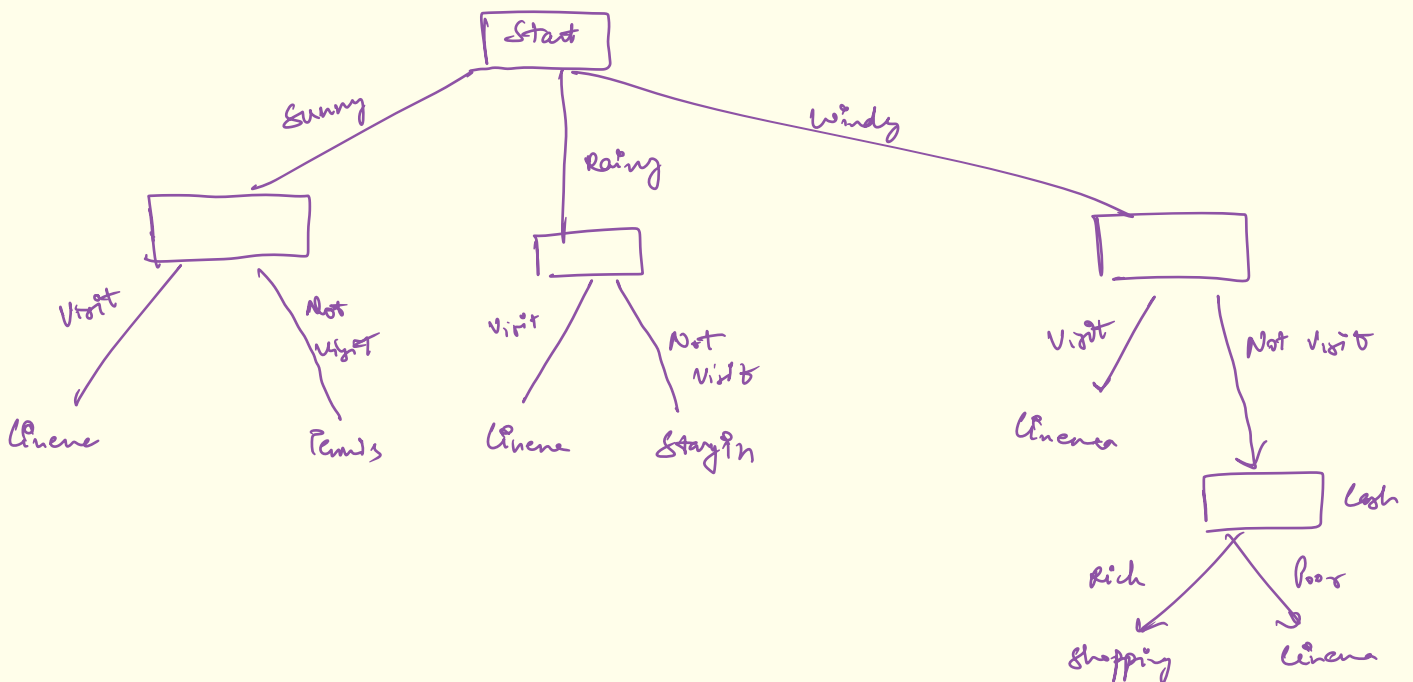
$$H(S_{\text{rich}}) = -\frac{2}{3} \log \frac{2}{3} - \frac{1}{3} \log \frac{1}{3} = 0.92$$

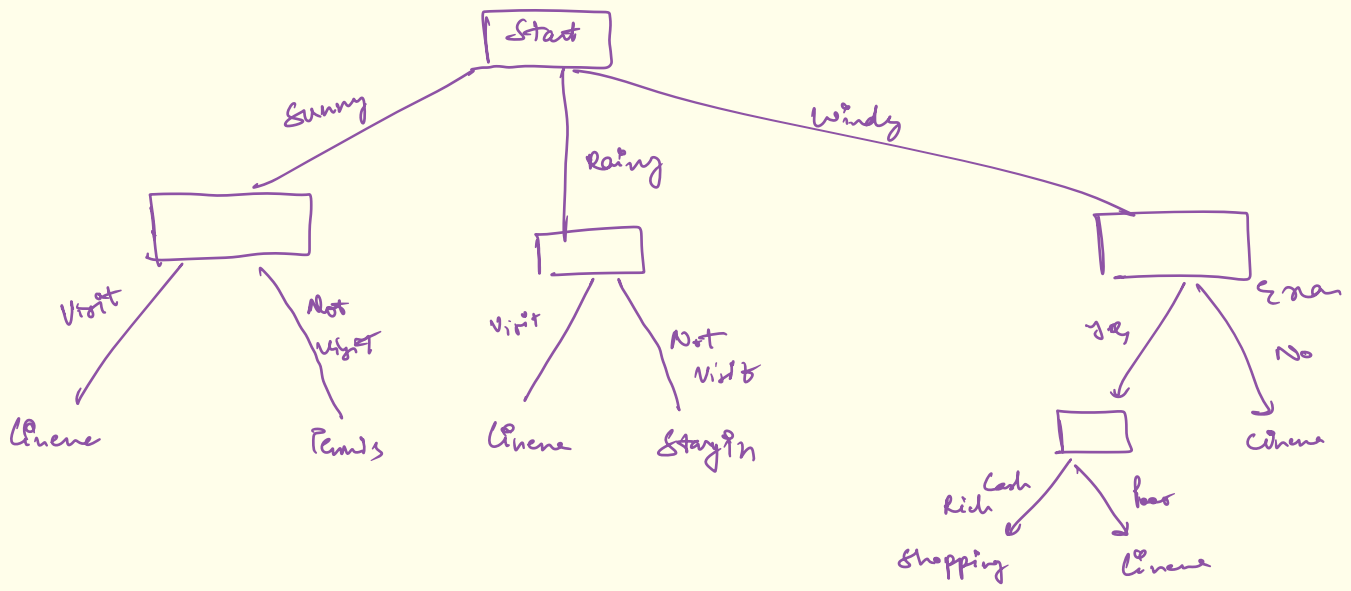
$$H(S_{\text{poor}}) = 0$$

$$H(S_{\text{yes}}) = -\frac{1}{2} \log \frac{1}{2} - \frac{1}{2} \log \frac{1}{2} = 1$$

$$H(S_{\text{no}}) = -\frac{2}{2} \log \frac{2}{2} - 0 - 0 = 0$$

Expected Tree Diagram





* Any of these works for final tree diagram-

Marking Scheme

4 marks for ID3 algorithm idea

4 marks for calculation.

2 marks for decision tree.